

Lista 1 - L1

①  $f(x) = x - x \cdot \ln(x)$  e  $\epsilon = 10^{-3}$

②  $[2,7; 2,8] \rightarrow$  Bisseção

| i | a      | b     | $\bar{x}$ | $f(a)$ | $f(b)$ | $f(\bar{x})$ | $ b-a $ |
|---|--------|-------|-----------|--------|--------|--------------|---------|
| 0 | 2,7    | 2,8   | 2,75      | ⊕      | ⊖      | -0,0319      | 0,1     |
| 1 | 2,7    | 2,75  | 2,725     | ⊕      | ⊖      | -0,0067      | 0,05    |
| 2 | 2,7    | 2,725 | 2,7125    | ⊕      | ⊖      | 0,0058       | 0,0375  |
| 3 | 2,7125 | 2,725 | 2,71825   | ⊕      | ⊖      | -0,0004      | 0,0125  |

$\hookrightarrow \bar{x} \approx 2,7182$

③  $[1; 3,5]$

| i | a         | b         | $\bar{x}$ | $f(a)$ | $f(b)$ | $f(\bar{x})$ |
|---|-----------|-----------|-----------|--------|--------|--------------|
| 0 | 1         | 3,5       | 2,688     | ⊕      | ⊖      | 0,5          |
| 1 | $\bar{x}$ | 3,5       | 2,729     | ⊕      | ⊖      | -0,019       |
| 2 | 2,688     | $\bar{x}$ | 2,717     | ⊕      | ⊖      | -0,0007      |

$\hookrightarrow \bar{x} \approx 2,7172$

$$x_* = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

3)  $\epsilon = 10^{-5}$

a)  $\frac{x}{4} - \lg(x) = 0$  com  $x_0 = 1,2$   $f'(x) = \frac{1}{4} - \frac{1}{x \ln 10}$

| i | $x_n$  | $f(x_n)$ | $x_{n+1}$ | $f'(x_n)$ |
|---|--------|----------|-----------|-----------|
| 0 | 1,2    | -2,2705  | 0,6798    | -4,3669   |
| 1 | 0,8915 | -1,0155  | 0,4468    | -2,2836   |
| 2 | 0,4468 | 0,3674   | 0,0717    | -0,9795   |
| 3 | 0,0717 | 0,0538   | 0,0003    | -0,7551   |
| 4 |        |          |           |           |

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$\bar{x} = 0,0003$ ,  $\therefore f(\bar{x}) < \epsilon$

b)  $2x - \ln(x) + 4 = 0$  com  $x_0 = -0,5$

| i | $x_n$   | $f(x_n)$  | $x_{n+1}$ | $f'(x_n)$ |
|---|---------|-----------|-----------|-----------|
| 0 | -0,5    | 3,4792    | -2,599    | -1,12245  |
| 1 | -2,599  | -2,7560   | -2,650    | 2,903     |
| 2 | -2,650  | -0,8279   | -2,362    | 2,8815    |
| 3 | -2,362  | -0,021010 | -2,345    | 4,2888    |
| 4 | -2,345  | 0,0230    | -2,3542   | 2,69964   |
| 5 | -2,3542 | -0,91940  | -2,01439  | 2,70569   |

Err = -0,00851

$f(x_{n+1}) < \epsilon \therefore \bar{x} = -2,3542$

③  $3x^4 - x = 3$  com  $x_0 = 1,5$

$f'(x) = 12x^3 - 1$

$f(x) = 3x^4 - x - 3 = 0$

| i | $x_n$  | $f(x_n)$ | $x_{n+1}$ | $f'(x_n)$ |
|---|--------|----------|-----------|-----------|
| 0 | 1,5    | 10,6875  | 1,2294    | 39,5      |
| 1 | 1,2294 | 2,6238   | 1,1062    | 21,2977   |
| 2 | 1,1062 | 0,3859   | 1,0808    | 15,2435   |
| 3 | 1,0808 | 0,0127   | 1,0799    | 14,1514   |
| 4 | 1,0799 | 0,0012   | 1,0799    | 13,7706   |

$> \text{Err} < 10^{-3}$

$\bar{x} = 1,0799$

④  $2\cos(x) - \frac{e^x}{2} = 0$  com  $1,9 = x_0$

| i | $x_n$  | $f(x_n)$ | $x_{n+1}$ | $f'(x_n)$ |
|---|--------|----------|-----------|-----------|
| 0 | 1,9    | -3,9895  | 1,1380    | -5,2355   |
| 1 | 1,1380 | -0,7214  | 0,9243    | -3,37585  |
| 2 | 0,9243 | -0,05526 | 0,9049    | -2,8564   |
| 3 | 0,9049 | -0,00313 | —         | —         |

$f'(x) = -2\sin(x) - \frac{1}{2}e^x$

$\bar{x} = 0,9049, \therefore f(\bar{x}) < \epsilon$

⑤  $p(x) = x^5 - \frac{10}{9}x^3 + \frac{5}{21}x$   $(-1; 1)$   $\epsilon = 10^{-4}$

⑥  $p'(x) = 5x^4 - \frac{10}{3}x^2 + \frac{5}{21}$   $x_0 = -0,85$

| i | $x_n$   | $f(x_n)$ | $f'(x_n)$ | $x_{n+1}$ |
|---|---------|----------|-----------|-----------|
| 0 | -0,85   | 0,0362   | 0,4397    | -0,9324   |
| 1 | -0,9324 | -0,0261  | 1,1200    | -0,9091   |
| 2 | -0,9091 | -0,0026  | 0,8988    | -0,9062   |
| 3 | -0,9062 | -0,00003 | —         | —         |

$> \text{Err} > \epsilon$

$f(x_n) < \epsilon \therefore \bar{x} = -0,9062$

⑥

| i | a         | b         | $\bar{x}$ | $f(a)$ | $f(b)$ | $f(\bar{x})$ | $ b-a $ |
|---|-----------|-----------|-----------|--------|--------|--------------|---------|
| 0 | -0,65     | -0,52     | -0,585    | ⊕      | ⊖      | 0,0146       | 0,1300  |
| 1 | -0,585    | -0,52     | -0,5525   | ⊕      | ⊖      | 0,0043       | +0,0325 |
| 2 | -0,5525   | -0,52     | -0,5362   | ⊕      | ⊖      | 0,0006       | —       |
| 3 | -0,5362   | $\bar{x}$ | -0,5443   | ⊕      | ⊖      | 0,0018       | —       |
| 4 | $\bar{x}$ | -0,5362   | -0,5403   | ⊕      | ⊖      | 0,0005       | —       |
| 5 | $\bar{x}$ | -0,5362   | -0,5382   | ⊕      | ⊖      | -0,0005      | —       |

$$\bar{x} = -0,5382$$

⑦

| i | a  | b | $\bar{x}$ | $f(a)$ | $f(b)$ | $f(\bar{x})$ | a       | b      |
|---|----|---|-----------|--------|--------|--------------|---------|--------|
| 0 | -0 |   | -0,0227   | ⊖      | ⊕      | -0,0053      | -0,3    | 0,25   |
|   |    |   | 0,0076    | ⊖      | ⊕      | 0,0018       | -0,0227 | 0,25   |
|   |    |   | -0,0000   | ⊖      | ⊕      | -0,0000      | -0,0227 | 0,0076 |

$$\bar{x} = -0,000012$$

④  $f(x) = x + 2\cos(x)$   $\epsilon = 10^{-2}$   $[-3; 0,5]$

| i | a     | b       | $\bar{x}$ | $f(a)$ | $f(b)$ | $f(\bar{x})$ | $ b-a $ |
|---|-------|---------|-----------|--------|--------|--------------|---------|
| 0 | -3    | 0,5     | -1,25     | ⊖      | ⊕      | -0,6193      | —       |
| 1 | -1,25 | 0,5     | -0,3750   | ⊖      | ⊕      | 1,4860       | —       |
| 2 | -1,25 | -0,3750 | -0,8125   | ⊖      | ⊕      | 0,5628       | —       |
| 3 | -1,25 | -0,8125 | -1,0312   | ⊖      | ⊕      | -0,0037      | —       |

$$f(\bar{x}) < \epsilon \therefore \bar{x} = -1,0312$$

⑤  $f(x) = e^x - 4x^2$   $\theta$   $(0; 1)$  Determine  $\theta$  com  $\epsilon = 10^{-2}$

②

| i | a      | b | $\bar{x}$ | $f(a)$   | $f(b)$    | $f(\bar{x})$ |
|---|--------|---|-----------|----------|-----------|--------------|
| 0 | 0      | 1 | 0.4382    | $\oplus$ | $\ominus$ | 0.7817       |
| 1 | 0.4382 | 1 | 0.6510    | $\oplus$ | $\ominus$ | 0.2220       |
| 2 | 0.6510 | 1 | 0.7025    | $\oplus$ | $\ominus$ | 0.0444       |
| 3 | 0.7025 | 1 | 0.7125    | $\oplus$ | $\ominus$ | 0.0082       |

$$\bar{x} = 0.7125, \quad f(\bar{x}) < \epsilon$$

⑥

| i | a | b | $\bar{x}$ | $f(a)$ | $f(b)$ | $f(\bar{x})$ |
|---|---|---|-----------|--------|--------|--------------|
| 0 |   |   |           |        |        |              |

| i | $f(x_n)$ | $x_{n+1}$ | $x_n$  | $f'(x_n)$ |
|---|----------|-----------|--------|-----------|
| 0 | -0.3821  | 0.7283    | 0.6    | -2.9778   |
| 1 | -0.050   | 0.7149    | 0.7283 | -3.7549   |
| 2 | -0.0005  |           | 0.7149 |           |

$$LRF = 0.7149$$

$$Err_L = 0.7149 - 0.7283 = -0.0134 \therefore Err_L < \epsilon$$

$$\bar{x} = 0.7149$$

⑥  $x_0 = 3,07$   $\epsilon = 10^{-5}$

①  $\text{sen}(x) = 0$

| i | $x_n$  | $x_{n+1}$         | $f(x)$  | $f'(x)$           | <del><math>f(x_{n+1})</math></del> |
|---|--------|-------------------|---------|-------------------|------------------------------------|
| 0 | 3,07   | 3,1417            | 0,0715  | -0,9974           |                                    |
| 1 | 3,1417 | <del>3,1415</del> | -0,0002 | <del>-0,999</del> |                                    |
|   |        | 3,1415            |         | -1,000            |                                    |
| 2 | 3,1415 | —                 | 0,0000  |                   |                                    |

$f(\bar{x}) < \epsilon$ ,  $\therefore \bar{x} = 3,1415$

⑦

| n | $x_n$  | $x_{n+1}$ | $f(x)$ | $f'(x)$ | $f(x_{n+1})$ |
|---|--------|-----------|--------|---------|--------------|
| 0 | 3,07   | 3,1058    | 0,0025 | -0,7153 | 0,0006       |
| 1 | 3,1058 | 3,1237    | 0,0006 | -0,0357 | 0,0002       |
| 2 | 3,1237 | 3,1326    | 0,0001 | -0,0178 | 0,0004       |
|   |        |           |        |         | 0,00001      |
| 3 | 3,1326 | 3,1371    | 0,0004 | -0,0089 |              |

$f(x_{n+1}) < \epsilon$ ,  $\therefore \bar{x} = 3,1371$

⑦  $\epsilon = 10^{-3}$  e intervalo  $[0,04; 0,06]$

PV = 321, E = 91,50, P = 12, PM = 27

$$\frac{1 - (1+j)^{-12}}{j} = \frac{321 - 91,50}{27} \Rightarrow \frac{1 - (1+j)^{-12}}{j} = 8,5$$

$$\therefore f(j) = \frac{1 - (1+j)^{-12}}{j} - 8,5$$

| i | a      | b    | $\bar{x}$ | $f(a)$   | $f(b)$    | $f(\bar{x})$ | $ b-a $ |
|---|--------|------|-----------|----------|-----------|--------------|---------|
| 0 | 0.04   | 0.06 | 0.05      | $\oplus$ | $\ominus$ | 0.3632       | 0.02    |
| 1 | 0.05   | 0.06 | 0.0550    | $\oplus$ | $\ominus$ | 0.1185       | 0.01    |
| 2 | 0.0550 | 0.06 | 0.0575    | $\oplus$ | $\ominus$ | -0.00004     | 0.005   |

$$f(\bar{x}) < \varepsilon \therefore \underline{\bar{x} = 0.0575}$$

⑧  $\varepsilon = 10^{-3}$  e  $x_0 = 7$ ,  $v_0 = 28 \text{ m/s}$ ,  $s_0 = 5 \text{ m}$ ,  $g = -9.81$

$$h(t) = s_0 + v_0 t + \frac{1}{2} g t^2 \Rightarrow h(t) = 5 + 28t + \frac{1}{2} \cdot (-9.81) t^2$$

$$\therefore h(t) = 5 + 28t - 4.905 t^2 \quad h'(t) = 28 - 9.81t$$

| i | $x_n$  | $x_{n+1}$ | $f(x_n)$ | $f'(x_n)$ | $f(x_{n+1})$ |
|---|--------|-----------|----------|-----------|--------------|
| 0 | 7      | 6.0325    | -39.3450 | -40.6700  | -4.5906      |
| 1 | 6.0325 | 5.8853    | -4.5906  | -31.1796  | -0.106325    |
| 2 | 5.8853 | 5.8818    | -0.1063  | -29.7352  | -0.00006     |

$$f(x_{n+1}) < \varepsilon \therefore \underline{x = 5.8818}$$